

# iSAQB Module: Formal Methods - Overview

**Active Group GmbH**

Created: 2026-10-04 Sun 12:48

# Hello! My name is ...

- Who am I?
- What is my work/knowledge background?
- What do I expect?
- Miscellaneous?

# Active Group GmbH



- Tübingen, Germany
- software project development using functional programming and functional software architecture
- Scala, Clojure, F#, Haskell, OCaml, Erlang, Elixir, Swift, ...
- consulting
- trainings and coaching

Blog: <https://funktionale-programmierung.de>

Conference: <https://bobkonf.de>

# iSAQB

## International Software Architecture Qualification Board



## International Software Architecture Qualification Board

"The purpose of the association is to standardise the training of software architects internationally."

- founded in 2008
- develops standard curricula
- foundation level
- advanced level
- product and technology-neutral

# Module: Formal Methods

- Logic
- Specification and Implementation
- Formal Methods and the Development Process
- Tools
- Examples

| technological | methodical | communicative |
|---------------|------------|---------------|
| 20            | 10         | 0             |

# Goals

- architects know a collection of various different formal methods
- architects can design architectures, that are suitable for verification
- architects understand the consequence of using formal methods

# Together

- enough space for questions
- there are no stupid questions
- ask immediately if something is unclear
- exercises in groups
- teamwork in solving tasks

# Unacceptable Conduct

- harassment, including derogatory comments about:
  - gender
  - sexual orientation
  - descent
  - religion
  - disability
- intimidation
- defamation and reputational damage
- including "jokes"

# Possible Measures

- remark (brrrr ... **awkard**)
- formal warning
- exclusion (without refund)

# Formal Methods: Specification and Verification

## iSAQB Advanced Level Curriculum - Formal Methods (FM)

We'll answer the following questions:

- What can formal methods do?
- What is and what isn't a specification?
- Where should formal methods be applied?
- What are logical foundations for applying formal methods?
- What practical formal-methods tools exist?
- What are architectural implications and consequences of using formal methods?

# What this Training Won't

- Won't make you an expert logician.
- Won't make you proficient at using specific tools.
- Won't tell you concretely how to reach a specific safety or security certification using formal methods.
- Won't cover every single learning goal in depth.

# Structure

- Introduction
- Introduction to Logic
- Specifications as Code

... and then introductions to specific tools.

- Logic Appendix